

Lecture-18

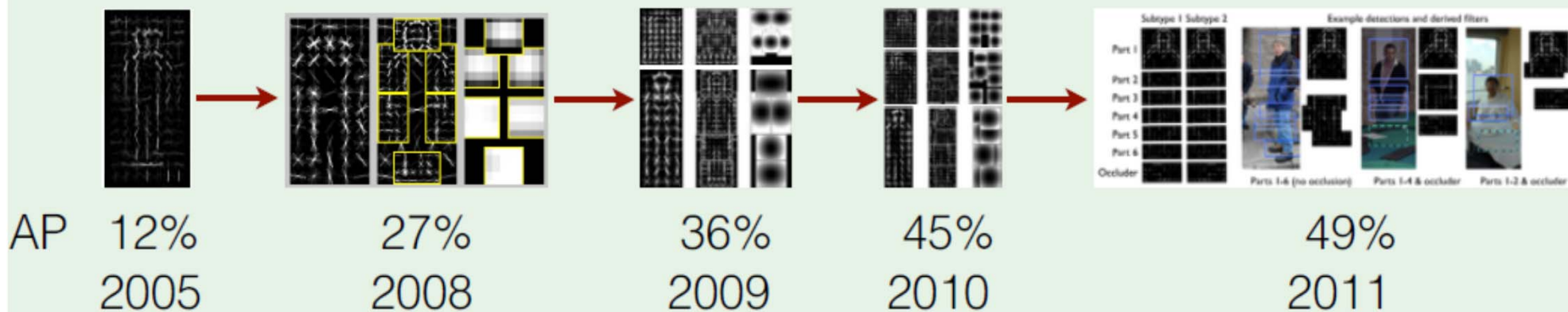
Deformable Part Model (DPM)

Object Detection with Deformable Part Models

[Cited by 2333](#)

(Slides Credits: Girshick, R. B, Yicong Tian and many others)

Felzenszwalb, P. F., Girshick, R. B., McAllester, D., & Ramanan, D. *Object detection with discriminatively trained part-based models*. In PAMI 2010. <http://www.cs.berkeley.edu/~rbg/latent/>



Person detection performance on PASCAL VOC 2007

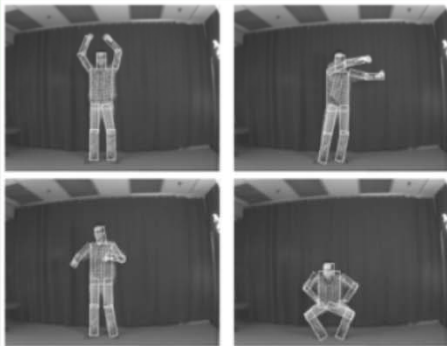
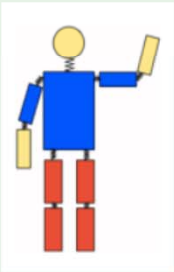
Slide Credit: ., Girshick, R. B

The Origins of DPM



Dalal & Triggs '05

- Histogram of Oriented Gradients (HOG)
- SVM training
- Sliding window detection



Fischler & Elschlager '73

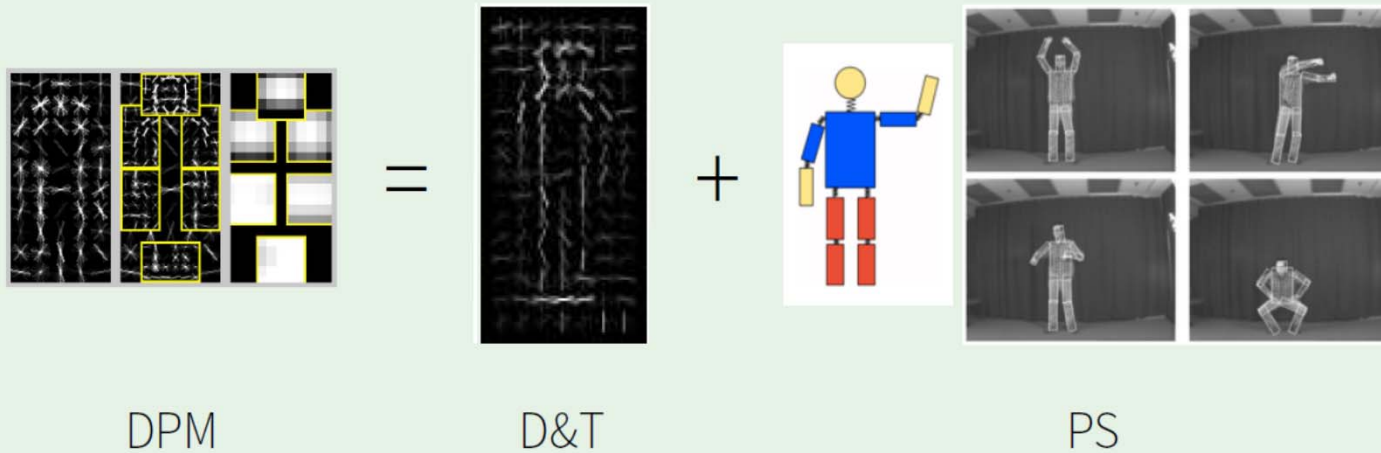
Felzenszwalb & Huttenlocher '00

- Pictorial structures
- Weak appearance models
- Non-discriminative training

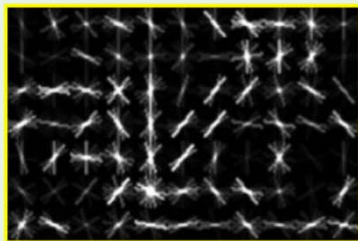
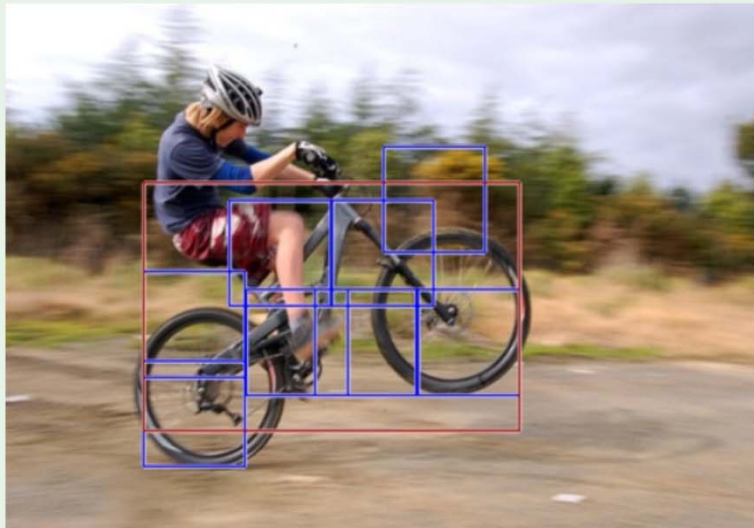
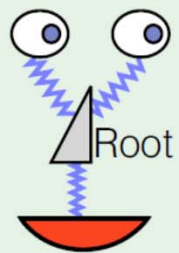
Slide Credit: ., Girshick, R. B

DPM key idea

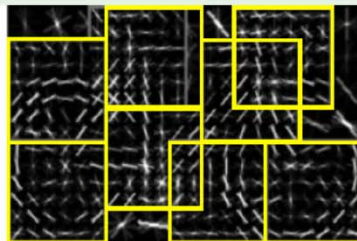
Port the success of Dalal & Triggs
into a part-based model



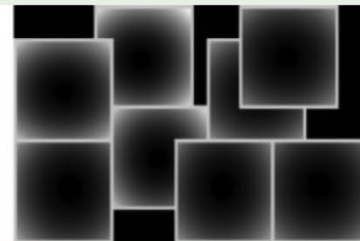
Example DPM



Root filter

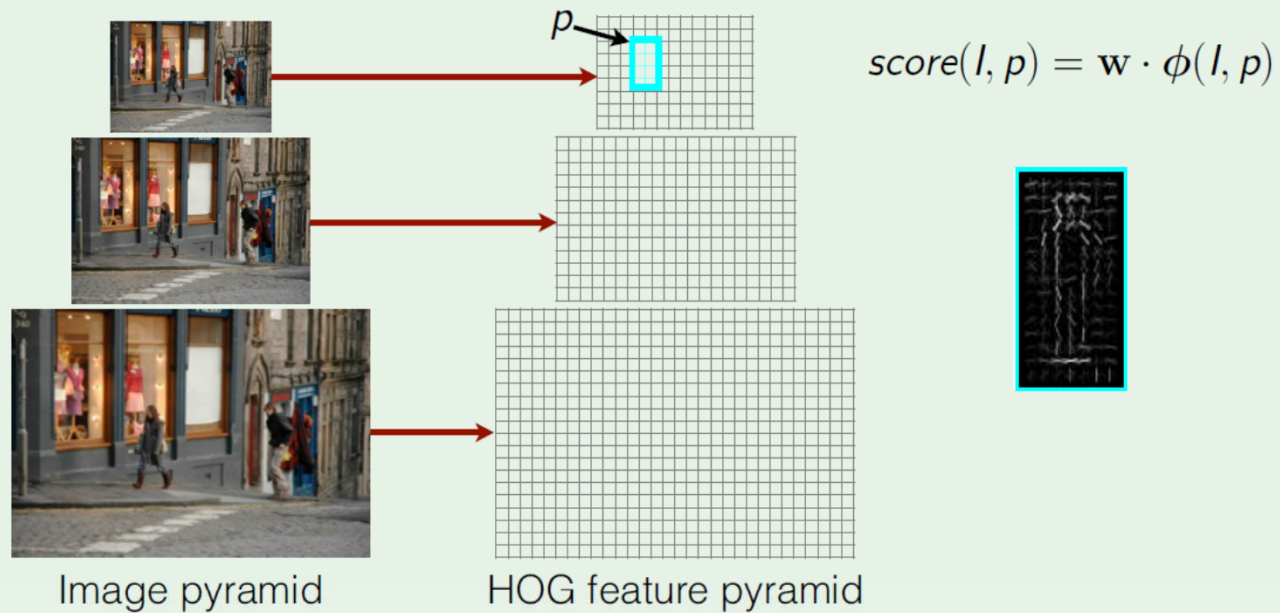


Part filters



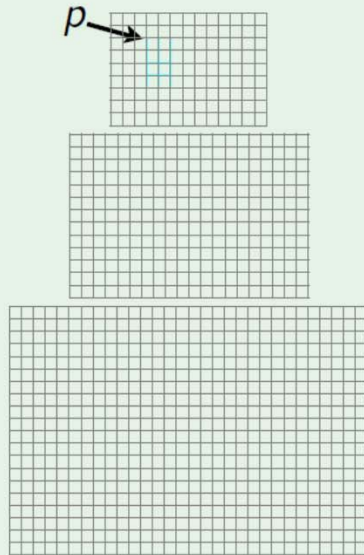
Deformation
costs

The Dalal & Triggs detector



- Compute HOG of the whole image at multiple resolutions
- Score every subwindow of the feature pyramid
- Apply non-maximal suppression

Detection



number of locations $p \sim 250,000$ per image

test set has ~ 5000 images

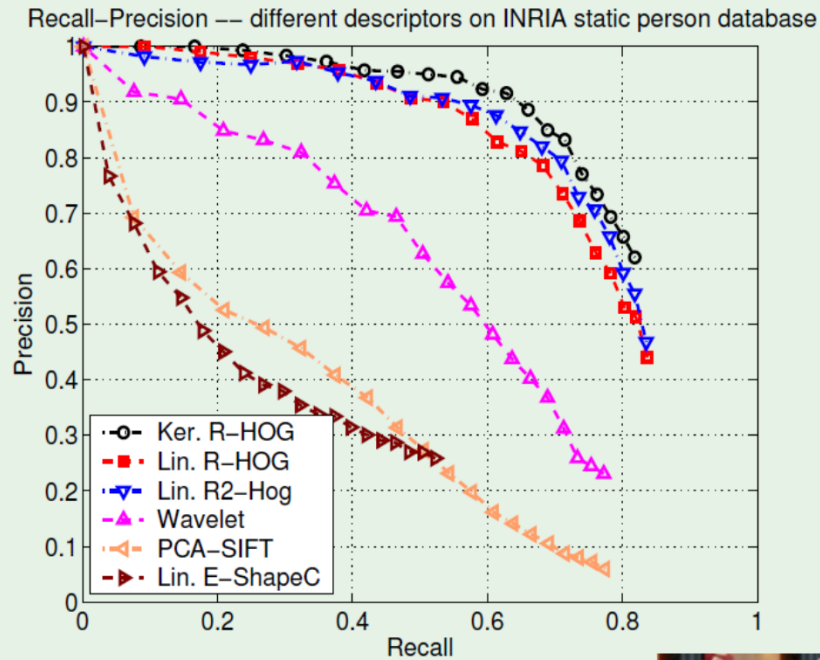
$\gg 1.3 \times 10^9$ windows to classify

typically only $\sim 1,000$ true positive locations

Extremely unbalanced binary classification

Slide Credit: ., Girshick, R. B

Dalal & Triggs detector on INRIA pedestrians



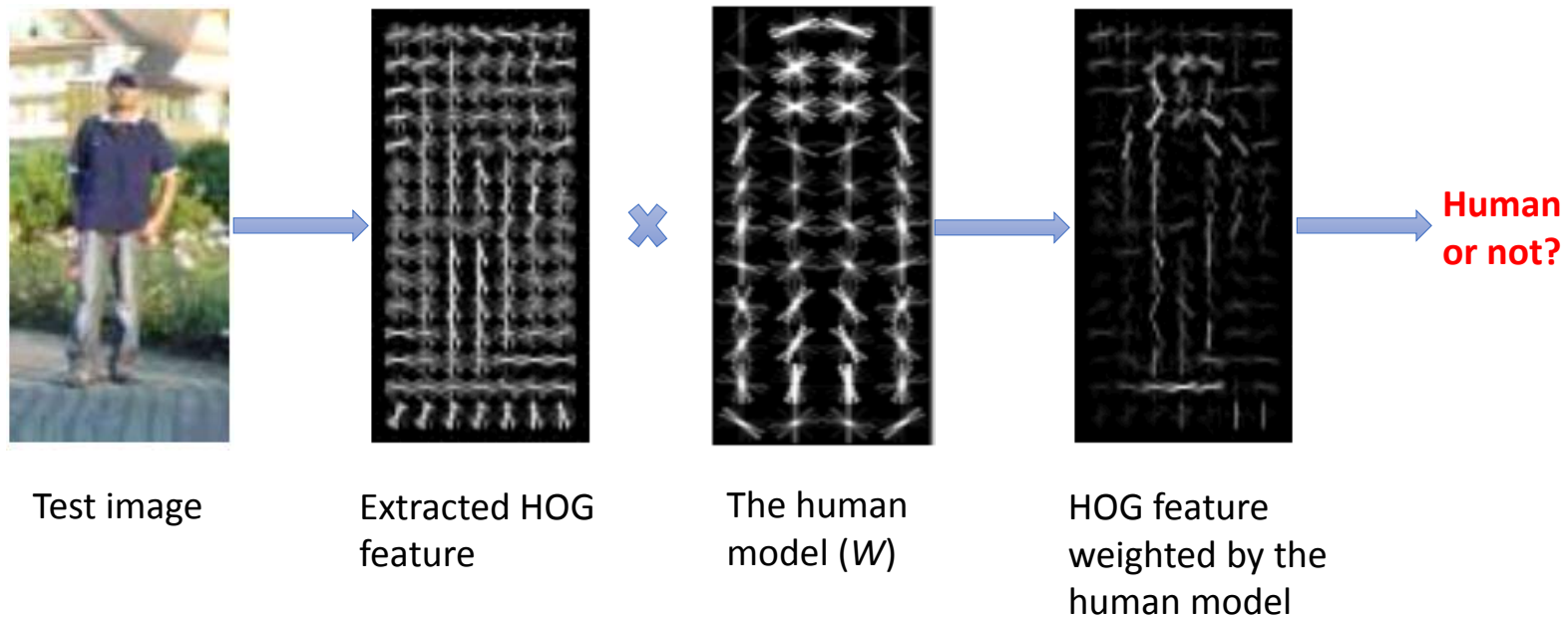
- AP = 75%
- (79% in my implementation)
- Very good
- **Declare victory and go home?**



Slide Credit: ., Girshick, R. B

Object Detection with HOG

- Dalal & Triggs detector: HOG + linear SVM

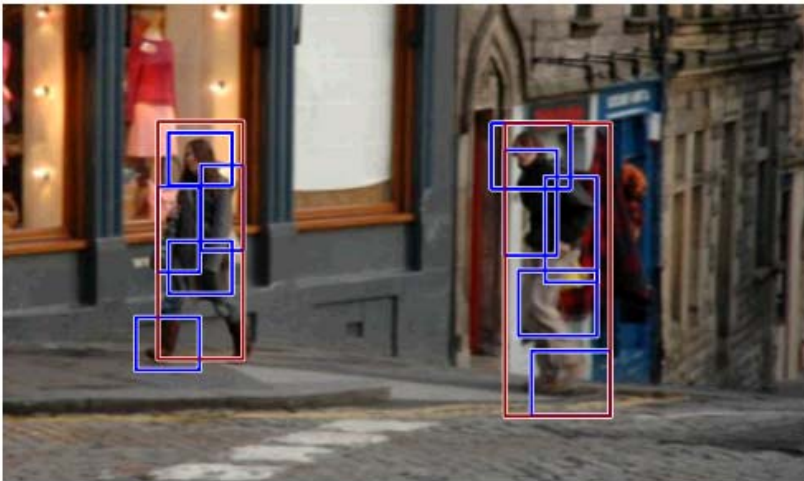


Dalal, N., & Triggs, B. *Histograms of oriented gradients for human detection*. In CVPR 2005.

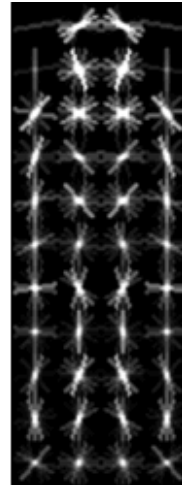
Why DPM?

- Great variations in objects
 - Non-rigid deformations, e.g. human in different poses
 - Intra-class variability, e.g. cars in various shape
 - Variations caused by different viewpoints and illumination
- A better representation of objects – deformable part model (DPM)
 - The Dalal & Triggs detector acts as the root filter in DPM

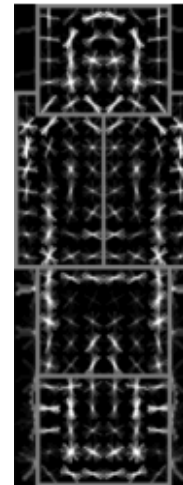
Overview



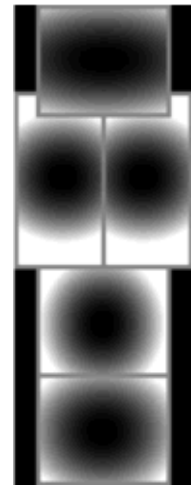
Example detection result



Coarse
root filter

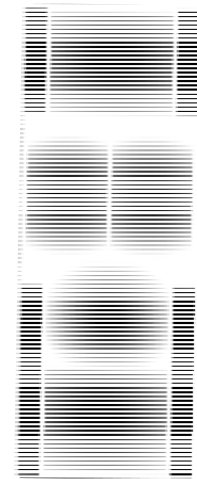
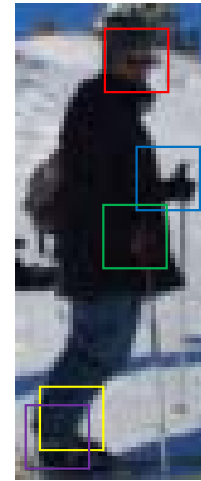
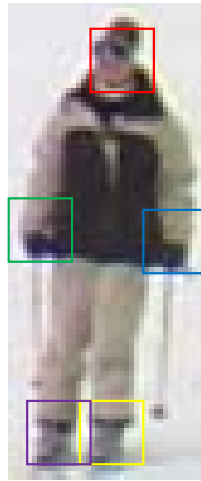
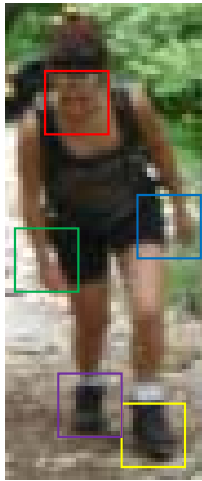


Higher resolution
part filters



Deformation
models

Starting Point

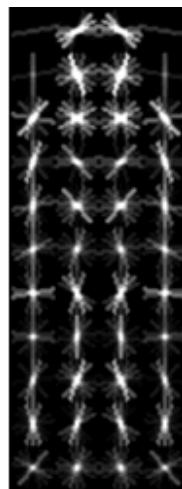


Deformation
models

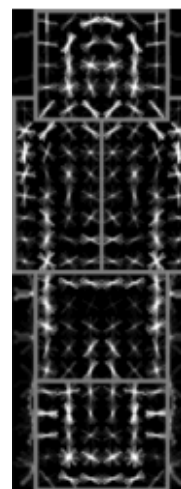
- Represent human by several parts
- The positions of the parts are not fixed, which means the model is deformable
- Penalize the parts that are far away from where they are supposed to be

Model Representation

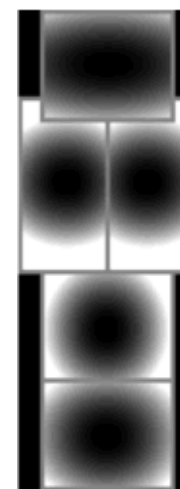
- A model has a root filter F_0 and n part models (F_i, v_i, d_i)
 - F_i : i -th part filter
 - v_i : anchor position of i -th part relative to the root
 - d_i : deformation parameters for i -th part



Coarse
root filter



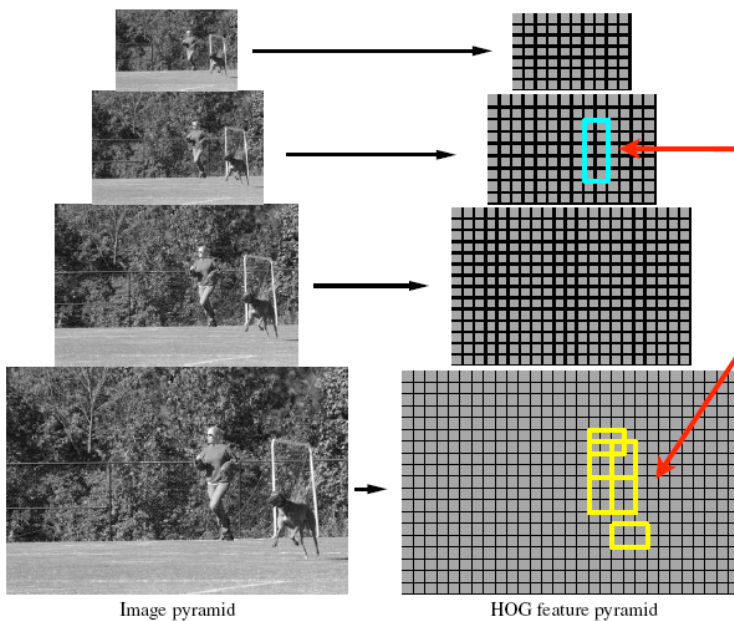
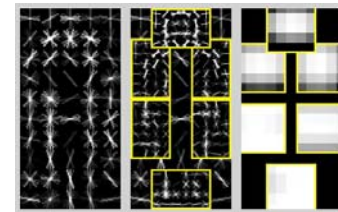
Higher resolution
part filters



Deformation
models

Object Hypothesis

- In HOG feature pyramid
 - root filter - coarser scale
 - part filters - finer scale



$$z = (p_0, \dots, p_n)$$

p_0 : location of root

$p_1, \dots, p_n$: location of parts

Score is sum of filter
scores minus
deformation costs

Score of a Hypothesis

$$score(p_0, \dots, p_n) = \sum_{i=0}^n F_i \cdot \phi(H, p_i) - \sum_{i=1}^n d_i \cdot \phi_d(dx_i, dy_i) + b$$

The diagram shows the equation for the score of a hypothesis. It consists of two yellow boxes: 'data term' and 'spatial prior'. The 'data term' box contains the sum $\sum_{i=0}^n F_i \cdot \phi(H, p_i)$. The 'spatial prior' box contains the sum $\sum_{i=1}^n d_i \cdot \phi_d(dx_i, dy_i)$. A blue arrow points from the label 'Filters' to F_i . Another blue arrow points from the label 'Feature of subwindow at location p_i ' to $\phi(H, p_i)$. A third blue arrow points from the label 'Deformation parameters' to d_i . A fourth blue arrow points from the label 'Displacement of part i relative to its anchor position' to $\phi_d(dx_i, dy_i)$. A fifth blue arrow points from the label 'Bias' to b .

Score of a hypothesis z is

$$score(z) = \beta \cdot \psi(H, z)$$

where

$$\beta = (F_0, \dots, F_n, d_1, \dots, d_n, b) \quad \text{Unknown}$$

$$\psi(H, z) = (\phi(H, p_0), \dots, \phi(H, p_n), -\phi_d(dx_1, dy_1), \dots, -\phi_d(dx_n, dy_n), 1) \quad \text{Known}$$

Score of a Hypothesis

$$score(p_0, \dots, p_n) = \sum_{i=0}^n F_i \cdot \phi(H, p_i) - \sum_{i=1}^n d_i \cdot \phi_d(dx_i, dy_i) + b$$

The equation is composed of three main parts:

- data term**: $\sum_{i=0}^n F_i \cdot \phi(H, p_i)$
 - F_i : Filters
 - $\phi(H, p_i)$: Feature of subwindow at location p_i
- spatial prior**: $-\sum_{i=1}^n d_i \cdot \phi_d(dx_i, dy_i)$
 - d_i : Deformation parameters
 - $\phi_d(dx_i, dy_i)$: Displacement of part i relative to its anchor position
- Bias**: $+ b$

The deformation function is defined as:

$$\phi_d(dx, dy) = (dx, dy, dx^2, dy^2)$$

The initial value for the deformation parameters is:

$$d_i = (0, 0, 1, 1)$$
 Initial Value (to be learned)

The diagram also shows a vertical stack of four grayscale images labeled **Deformation models**, which represent different spatial priors or deformation fields.

Detection

- The overall score of a root location is computed according to the best possible placement of the parts

$$score(p_0) = \max_{p_1, \dots, p_n} score(p_0, \dots, p_n)$$

- High-scoring root locations define detections
- Sliding-window approach
- Efficient computation ($O(nk)$): dynamic programming + generalized distance transforms

Detection

- Distance transform

- Response of the i -th part filter in the l -th level of the feature pyramid

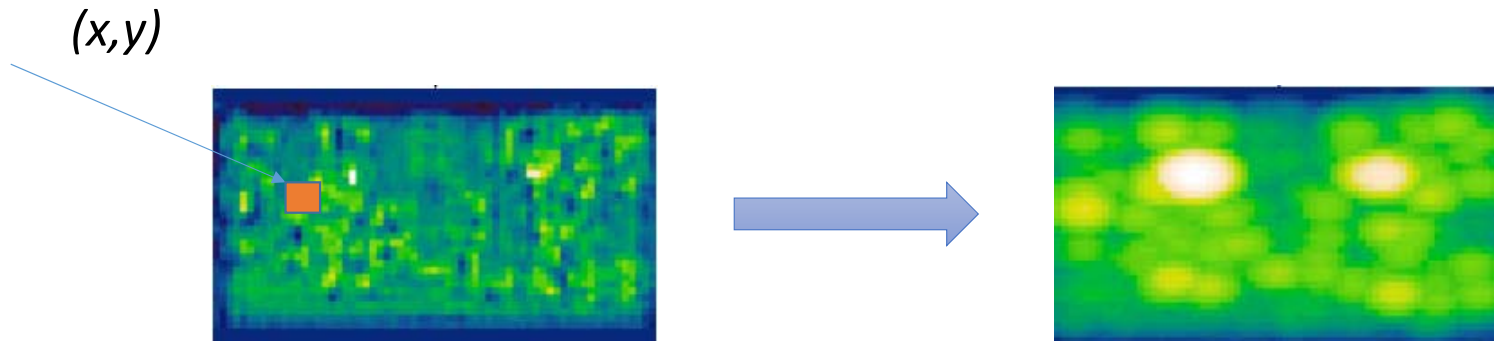
$$R_{i,l}(x, y) = F_i \cdot \phi(H, (x, y, l))$$

- Transformed response, given root is at (x, y)

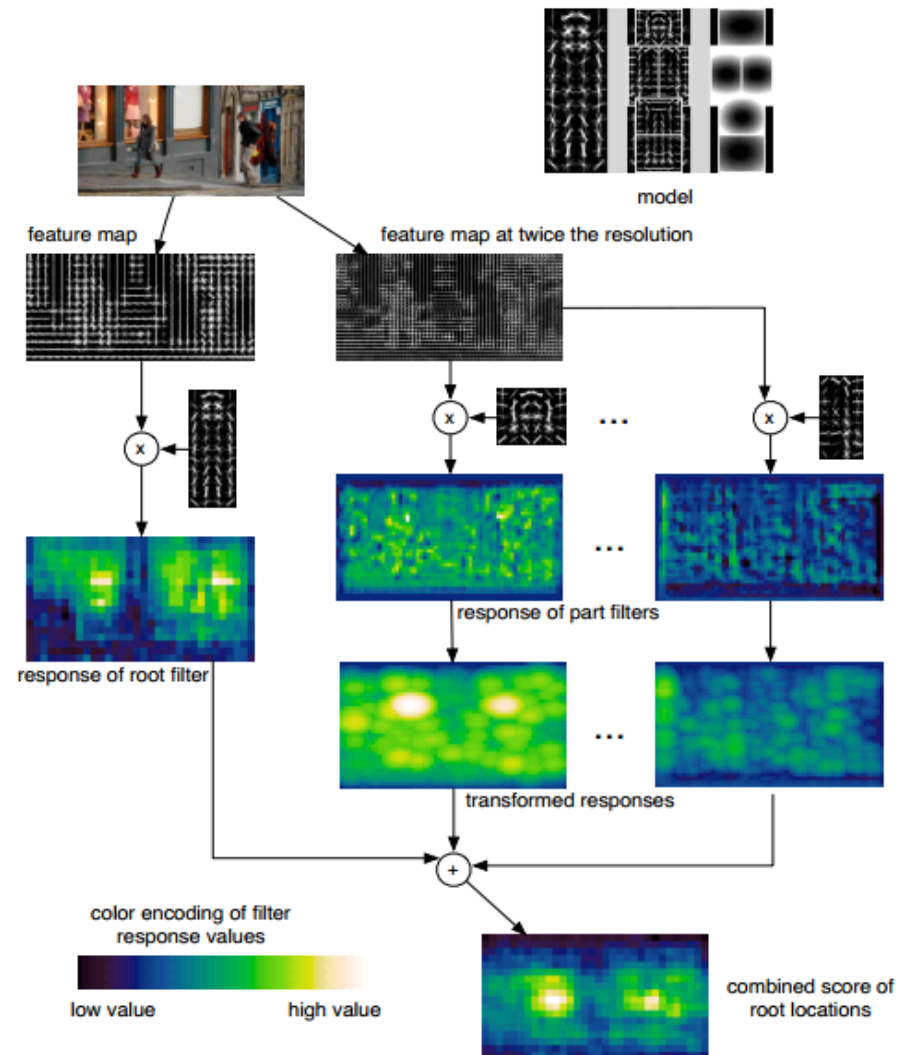
$$D_{i,l}(x, y) = \max_{dx, dy} (R_{i,l}(x + dx, y + dy) - d_i \cdot \phi_d(dx, dy))$$

$$\phi_d(dx, dy) = (dx, dy, dx^2, dy^2)$$

$$d_i = (0, 0, 1, 1)$$

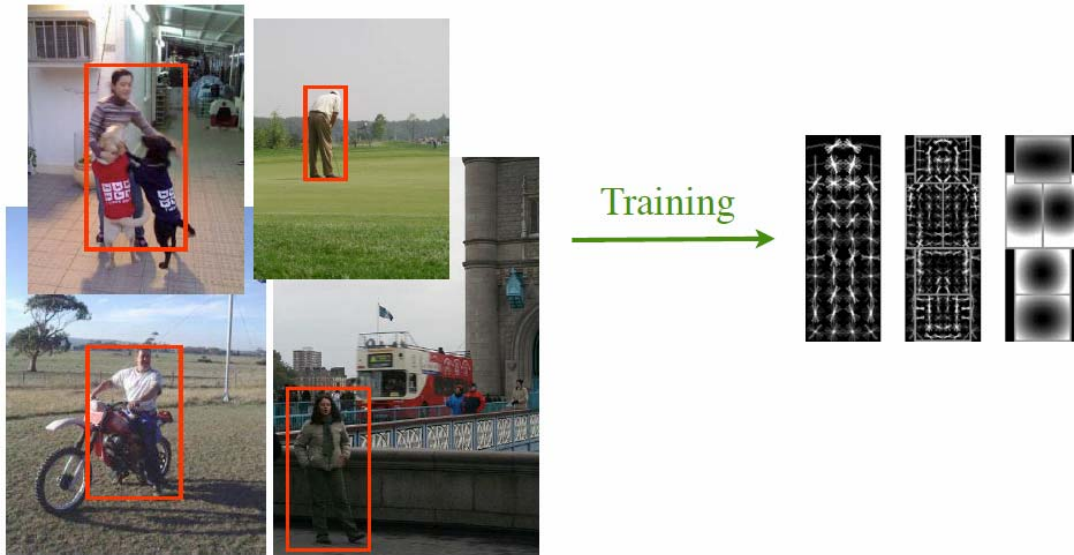


Detection

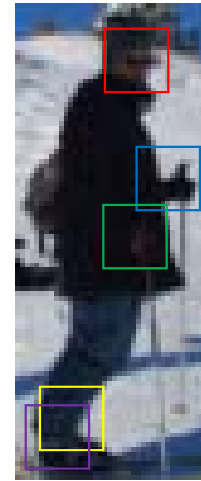
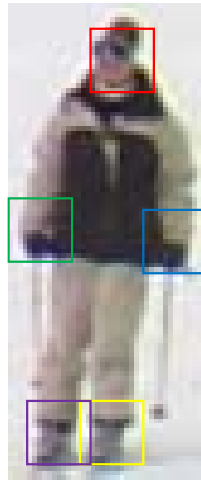
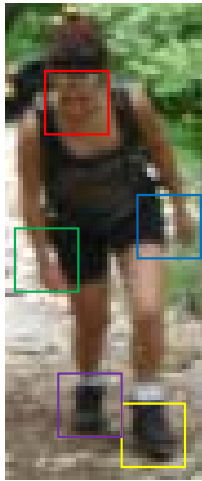


Training

- Positive training examples are labeled with bounding boxes
- No part location is available during training (latent)
- Aim: learn model parameters $\beta = (F_0, \dots, F_n, d_1, \dots, d_n, b)$



Latent Variables



- The positions of the parts are not given in both the training and the testing images
- The variables that exist but not known in training samples are called latent variables
- The learning algorithm must be able to find/discover the optimal values for the latent variables, namely the position of the parts.

Training

- The classifier scores an example x by

$$f_{\beta}(x) = \max_{z \in Z(x)} \beta \cdot \Phi(x, z)$$

- β : the model parameters
- z : latent values
- $Z(x)$: the possible latent values for example x

Training

- Minimize the objective function

$$L_D(\beta) = \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n \max(0, 1 - y_i f_\beta(x_i))$$

- Labeled training examples $D = (\langle x_1, y_1 \rangle, \dots, \langle x_n, y_n \rangle)$
- $y_i \in \{-1, 1\}$

Optimization

Learning an SVM has been formulated as a **constrained** optimization problem over $\mathbf{w}$ and ξ

$$\min_{\mathbf{w} \in \mathbb{R}^d, \xi_i \in \mathbb{R}^+} \|\mathbf{w}\|^2 + C \sum_i^N \xi_i \text{ subject to } y_i (\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i \text{ for } i = 1 \dots N$$

The constraint $y_i (\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i$, can be written more concisely as

$$y_i f(\mathbf{x}_i) \geq 1 - \xi_i$$

which, together with $\xi_i \geq 0$, is equivalent to

$$\xi_i = \max(0, 1 - y_i f(\mathbf{x}_i))$$

Hence the learning problem is equivalent to the **unconstrained** optimization problem over $\mathbf{w}$

$$\min_{\mathbf{w} \in \mathbb{R}^d} \underbrace{\|\mathbf{w}\|^2}_{\text{regularization}} + C \sum_i^N \underbrace{\max(0, 1 - y_i f(\mathbf{x}_i))}_{\text{loss function}}$$

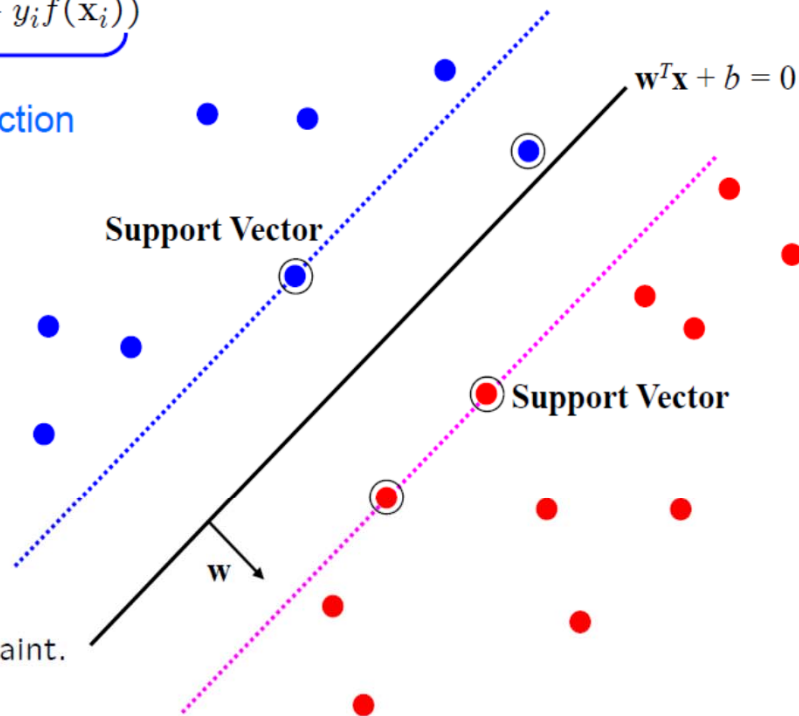
Slide Credit: A. Zisserman

Loss function

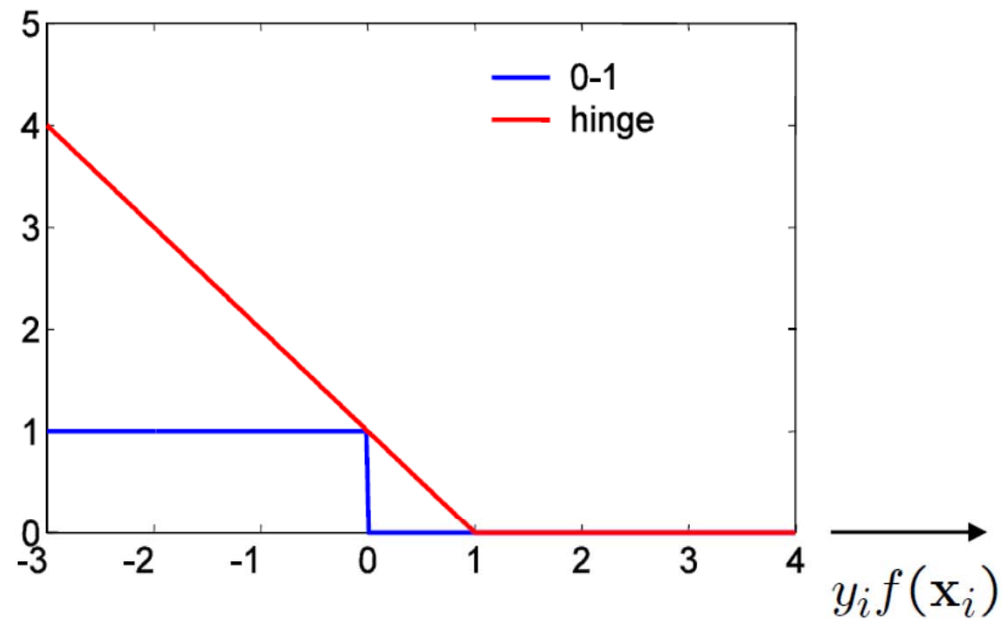
$$\min_{\mathbf{w} \in \mathbb{R}^d} \|\mathbf{w}\|^2 + C \sum_i \underbrace{\max(0, 1 - y_i f(\mathbf{x}_i))}_{\text{loss function}}$$

Points are in three categories:

1. $y_i f(\mathbf{x}_i) > 1$
Point is outside margin.
No contribution to loss
2. $y_i f(\mathbf{x}_i) = 1$
Point is on margin.
No contribution to loss.
As in hard margin case.
3. $y_i f(\mathbf{x}_i) < 1$
Point violates margin constraint.
Contributes to loss



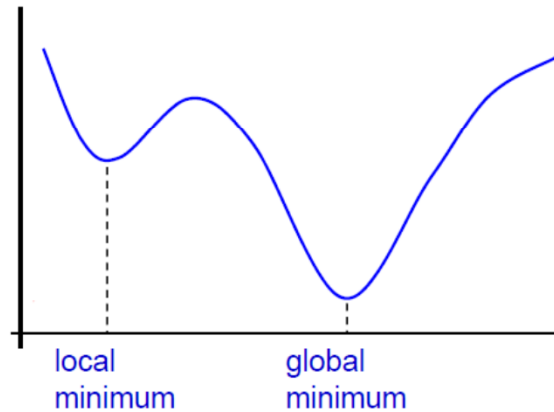
Loss functions



- SVM uses “hinge” loss $\max(0, 1 - y_i f(\mathbf{x}_i))$
- an approximation to the 0-1 loss

Optimization continued

$$\min_{\mathbf{w} \in \mathbb{R}^d} C \sum_i^N \max(0, 1 - y_i f(\mathbf{x}_i)) + \|\mathbf{w}\|^2$$



- Does this cost function have a unique solution?
- Does the solution depend on the starting point of an iterative optimization algorithm (such as gradient descent)?

If the cost function is **convex**, then a locally optimal point is globally optimal (provided the optimization is over a convex set, which it is in our case)

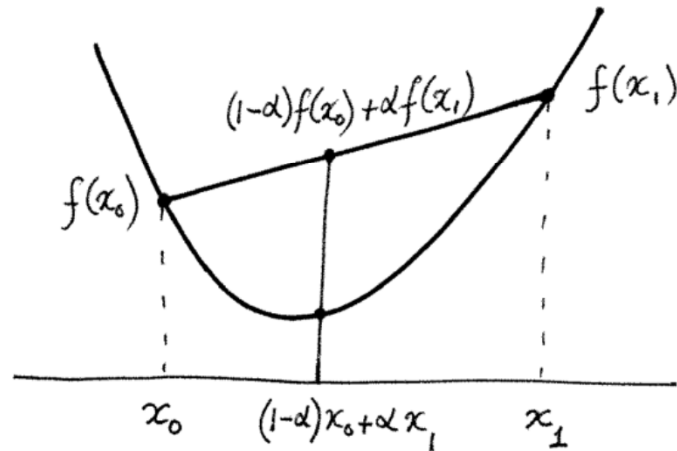
Convex functions

D – a domain in $\mathbb{R}^n$.

A **convex function** $f : D \rightarrow \mathbb{R}$ is one that satisfies, for any x_0 and x_1 in D :

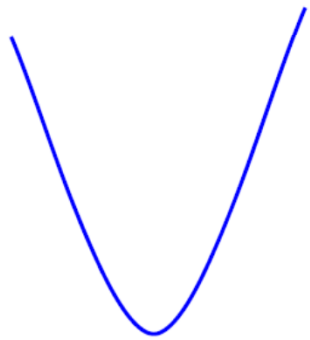
$$f((1 - \alpha)x_0 + \alpha x_1) \leq (1 - \alpha)f(x_0) + \alpha f(x_1) .$$

Line joining $(x_0, f(x_0))$
and $(x_1, f(x_1))$ lies
above the function graph.

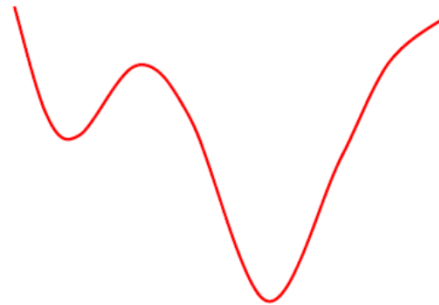
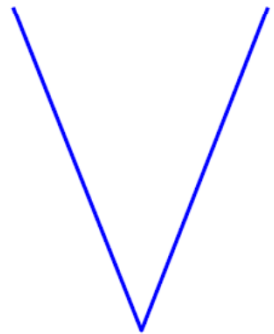


Slide Credit: A. Zisserman

Convex function examples



convex

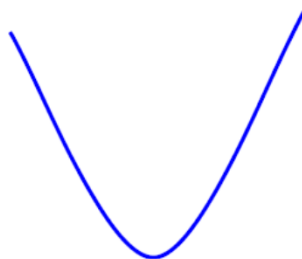


Not convex

A non-negative sum of convex functions is convex



+



SVM

$$\min_{\mathbf{w} \in \mathbb{R}^d} C \sum_i^N \max(0, 1 - y_i f(\mathbf{x}_i)) + \|\mathbf{w}\|^2$$

convex

Gradient (or steepest) descent algorithm for SVM

To minimize a cost function $\mathcal{C}(\mathbf{w})$ use the iterative update

$$\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta_t \nabla_{\mathbf{w}} \mathcal{C}(\mathbf{w}_t)$$

where η is the learning rate.

First, rewrite the optimization problem as an **average**

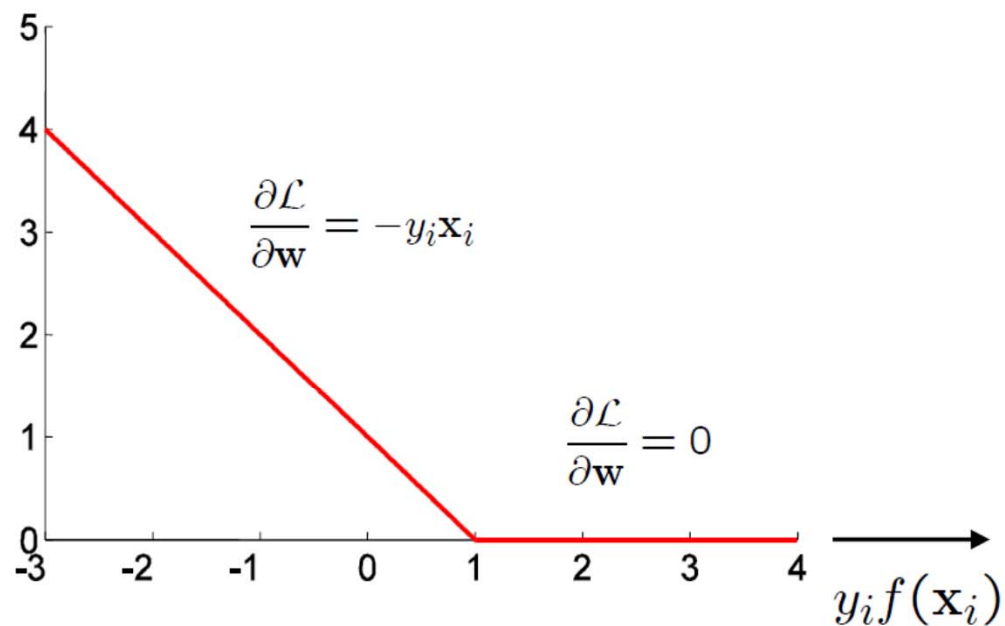
$$\begin{aligned} \min_{\mathbf{w}} \mathcal{C}(\mathbf{w}) &= \frac{\lambda}{2} \|\mathbf{w}\|^2 + \frac{1}{N} \sum_i^N \max(0, 1 - y_i f(\mathbf{x}_i)) \\ &= \frac{1}{N} \sum_i^N \left(\frac{\lambda}{2} \|\mathbf{w}\|^2 + \max(0, 1 - y_i f(\mathbf{x}_i)) \right) \end{aligned}$$

(with $\lambda = 2/(NC)$ up to an overall scale of the problem) and
 $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$

Because the hinge loss is not differentiable, a **sub-gradient** is computed

Sub-gradient for hinge loss

$$\mathcal{L}(\mathbf{x}_i, y_i; \mathbf{w}) = \max(0, 1 - y_i f(\mathbf{x}_i)) \quad f(\mathbf{x}_i) = \mathbf{w}^\top \mathbf{x}_i + b$$



Slide Credit: A. Zisserman

Sub-gradient descent algorithm for SVM

$$\mathcal{C}(\mathbf{w}) = \frac{1}{N} \sum_i^N \left(\frac{\lambda}{2} \|\mathbf{w}\|^2 + \mathcal{L}(\mathbf{x}_i, y_i; \mathbf{w}) \right)$$

The iterative update is

$$\begin{aligned} \mathbf{w}_{t+1} &\leftarrow \mathbf{w}_t - \eta \nabla_{\mathbf{w}_t} \mathcal{C}(\mathbf{w}_t) \\ &\leftarrow \mathbf{w}_t - \eta \frac{1}{N} \sum_i^N (\lambda \mathbf{w}_t + \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{x}_i, y_i; \mathbf{w}_t)) \end{aligned}$$

where η is the learning rate.

Then each iteration t involves cycling through the training data with the updates:

$$\begin{aligned} \mathbf{w}_{t+1} &\leftarrow \mathbf{w}_t - \eta(\lambda \mathbf{w}_t - y_i \mathbf{x}_i) && \text{if } y_i f(\mathbf{x}_i) < 1 \\ &\leftarrow \mathbf{w}_t - \eta \lambda \mathbf{w}_t && \text{otherwise} \end{aligned}$$

In the Pegasos algorithm the learning rate is set at $\eta_t = \frac{1}{\lambda t}$

Slide Credit: A. Zisserman

Latent SVM

- A latent SVM is semi-convex

- $f_{\beta}(x) = \max_{z \in Z(x)} \beta \cdot \Phi(x, z)$ is convex in β

- For negative examples ($y_i = -1$), the hinge loss is **convex**

$$\max(0, 1 - y_i f_{\beta}(x_i)) = \max(0, 1 + f_{\beta}(x_i))$$

(the maximum of two convex function)

- For positive examples ($y_i = 1$), the hinge loss is **not convex**

$$\max(0, 1 - y_i f_{\beta}(x_i)) = \max(0, 1 - f_{\beta}(x_i))$$

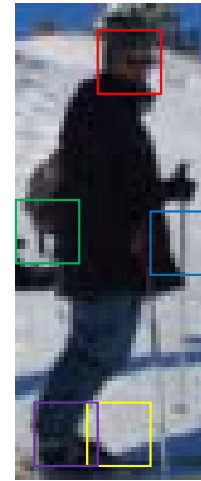
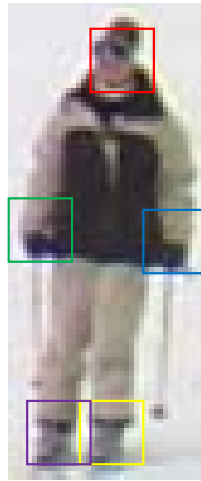
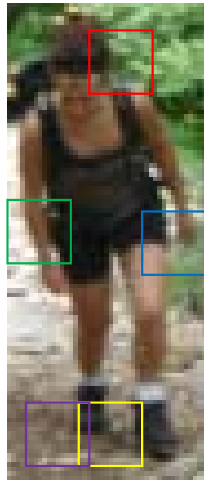
(the maximum of a convex function and a concave function)

- If the latent value for positive examples are fixed, the hinge loss is **convex**

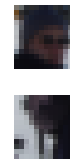
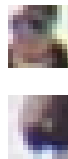
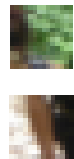
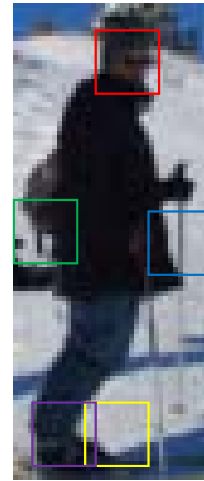
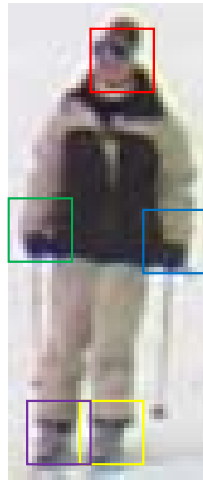
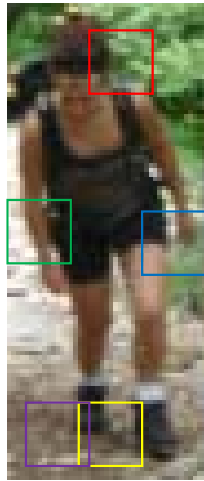
Latent SVM

- Initialize β using standard SVM by assuming the same parts locations for all the positive examples
- Iterative optimization:
 - Relabel positive examples: fix β , find the best z for each positive example (exactly the same with detection!)
 - Optimize β : fix z , optimize β by solving the convex problem

Training

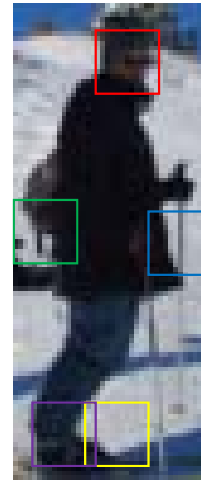
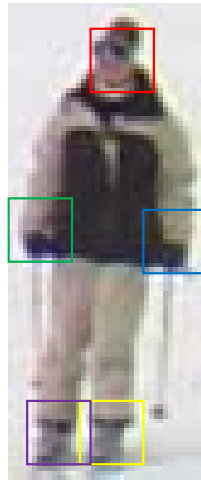
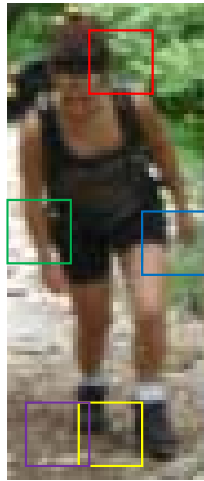


Step 1. we suppose the positions of the parts are fixed for each frame, and train part detector using SVM



Train SVM Classifier

Training



Step 2. Using the trained SVM classifier, find the optimal part position in each training sample

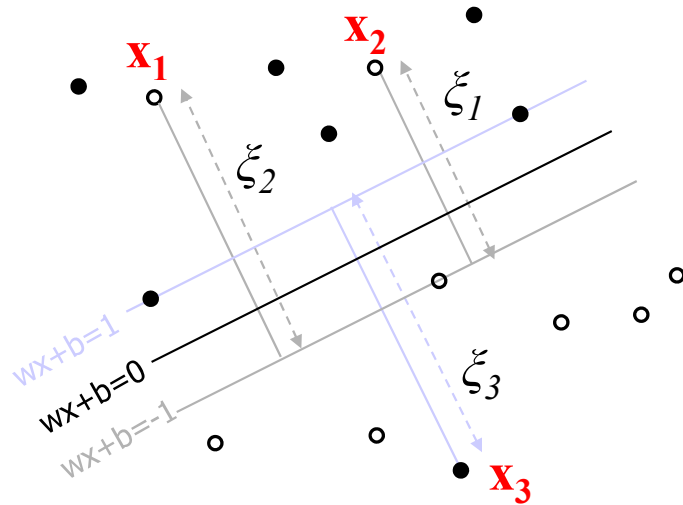
Negative examples

- A very large number of negative examples
 - Infeasible to consider all of them in training!
- Data-mining hard examples

$$H(\beta, D)$$

$$= \left\{ (i, \Phi(x_i, z_i)) \mid z_i = \operatorname{argmax}_{z \in Z(x_i)} \beta \cdot \Phi(x_i, z) \text{ and } y_i (\beta \cdot \Phi(x_i, z_i)) < 1 \right\}$$

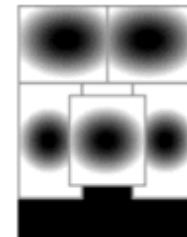
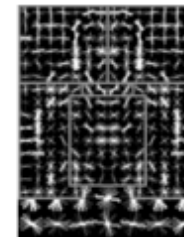
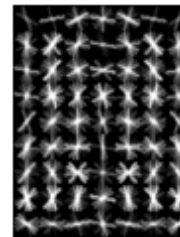
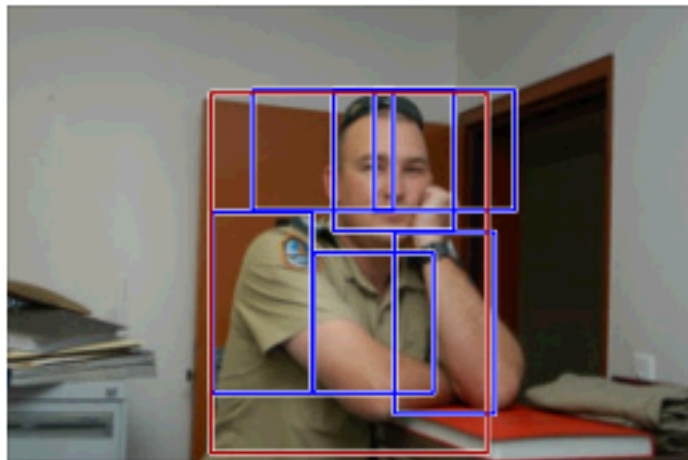
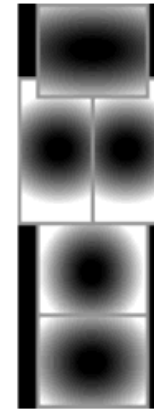
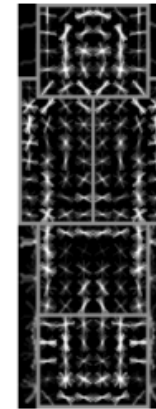
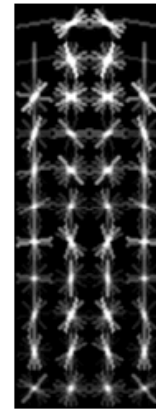
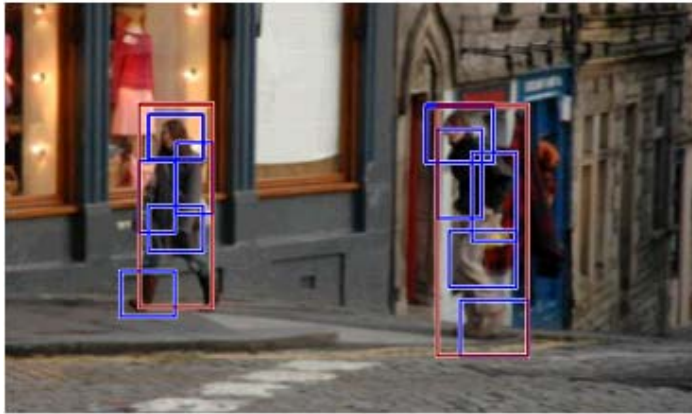
Hard Negatives



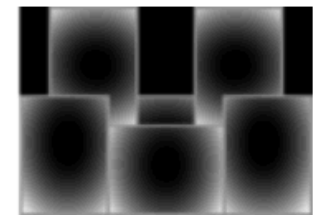
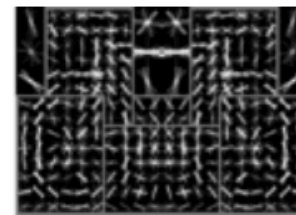
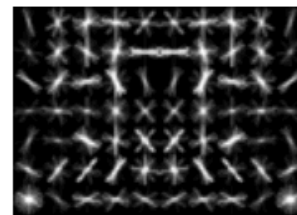
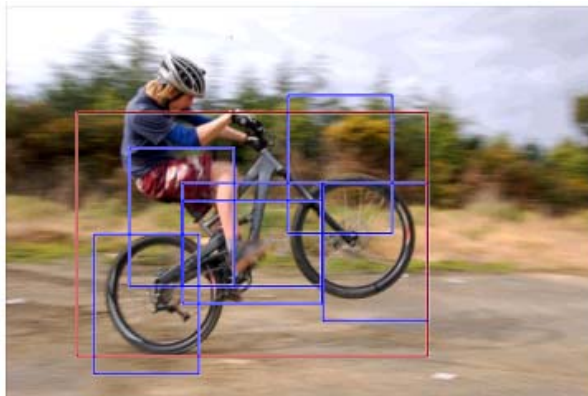
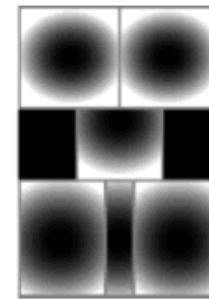
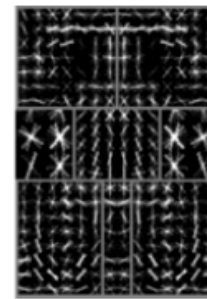
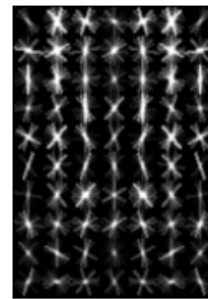
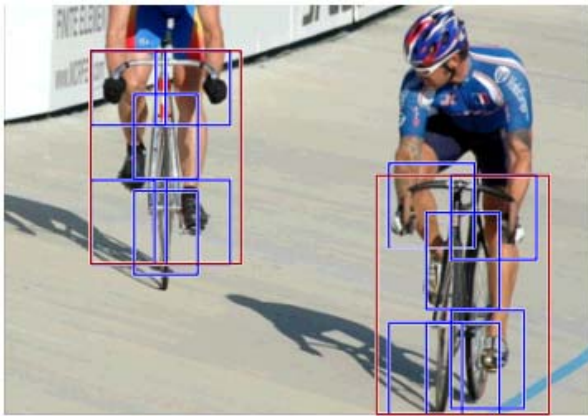
Mixture Model

- A mixture model consists of m components
- Captures extreme intra-class variation
- Split the positive bounding boxes into m groups by aspect ratio

Mixture Model Example - Person

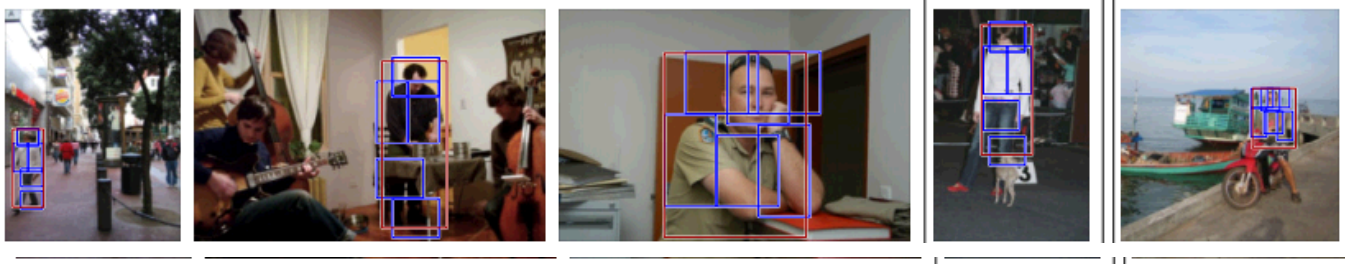


Mixture Model Example - Bicycle

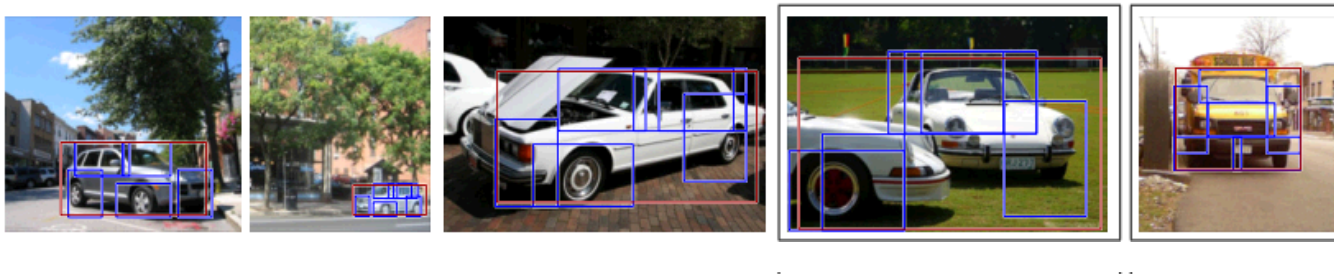


Qualitative Results (PASCAL 2007)

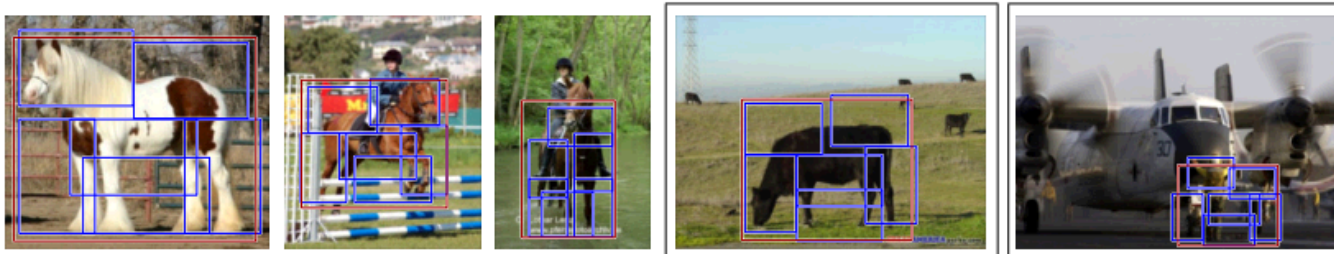
person



car

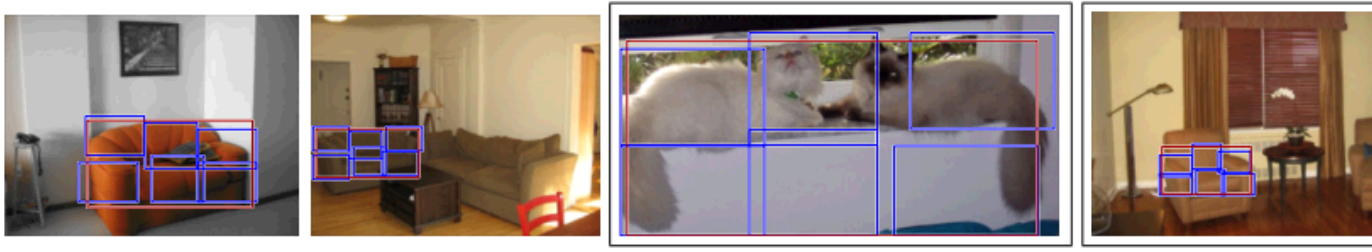


horse



Qualitative Results (PASCAL 2007)

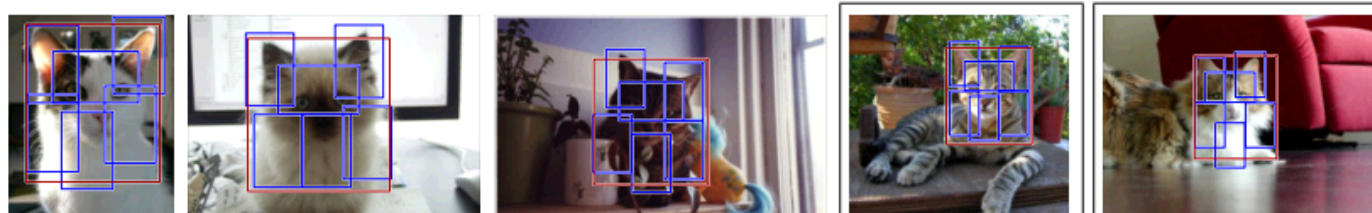
sofa



bottle



cat

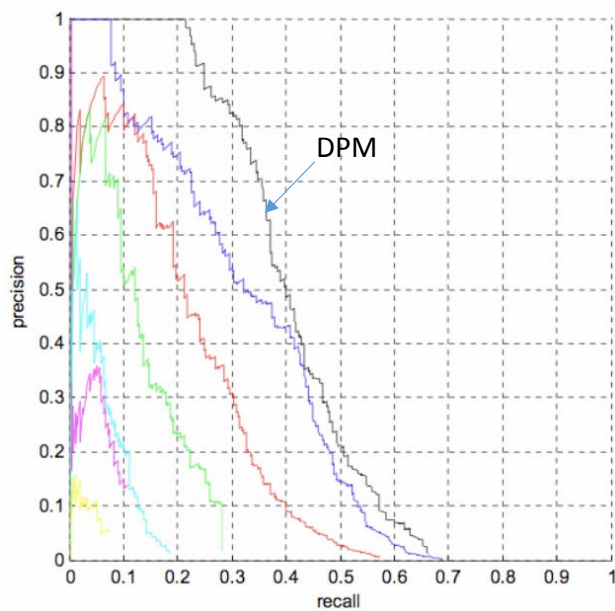


Quantitative Results (PASCAL 2008)

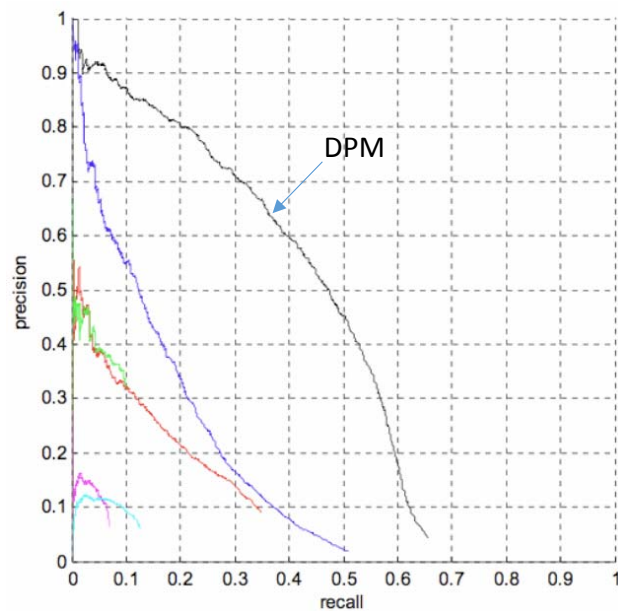
- 7 systems competed in the 2008 challenge
- Out of 20 classes, DPM got
 - First place in 7 classes
 - Second place in 8 classes

Quantitative Results (PASCAL 2008)

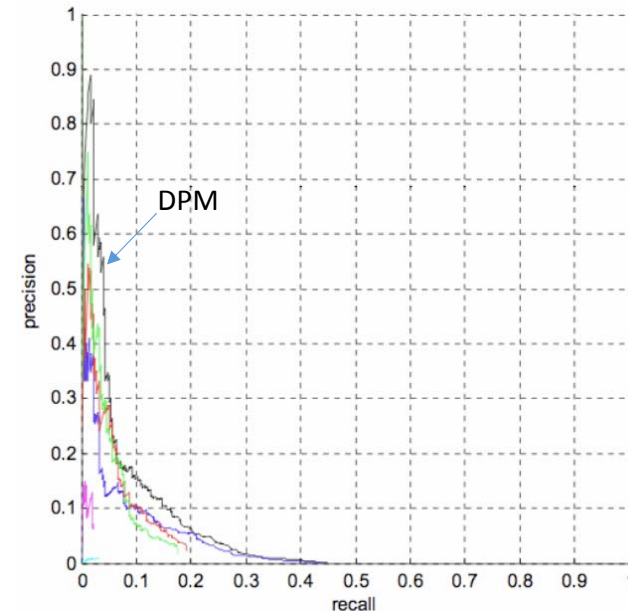
Bicycle



Person



Bird



Quantitative Results (PASCAL 2006)

- Comparison of different configurations of proposed model

